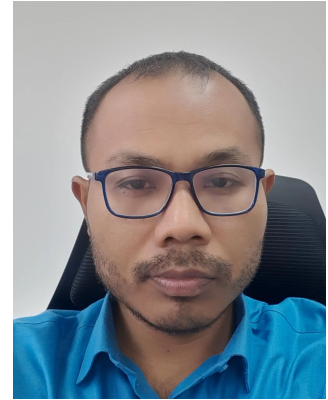


CURRICULUM VITAE



INDU KALPA DIHINGIA

CONTACT INFORMATION	1. Vill: Betanipam, Dhemaji Assam - 787059, India. 2. Building 8, No. 1308, Keyuan Road, Zhangjiang Science Park, Shanghai Jiaotong University, China.	<i>Mobile:</i> +91 8876925745 +86 15121018551 <i>E-mail(s):</i> ikd4638@gmail.com (Preferred) idihingia@iiti.ac.in ikd4638@sjtu.edu.cn (Mainland China) <i>Orcid ID:</i> 0000-0002-4064-0446
PERSONAL INFORMATION	<i>Date of Birth:</i> December 12, 1991 <i>Gender:</i> Male <i>Nationality:</i> Indian	
AREAS OF RESEARCH INTERESTS	Astrophysics and astronomy: Accretion flow in alternate theories of gravity and exotic objects, e.g., Naked Singularity (NkS). General relativistic magnetohydrodynamic (GRMHD) and ray tracing (GRRT) simulations. General relativistic magnetically driven flows from thin accretion disk. Accretion flow in single-temperature as well as in the two-temperature paradigm.	
MEMBERSHIP	Event horizon telescope collaboration (EHTC, since 2022).	
COMPUTATIONAL EXPERTISE	I have developed certain expertise on ◇ Programming with FORTRAN, C, JULIA and PYTHON. ◇ Analytic and numerical calculations using MATHEMATICA. ◇ Numerical simulation code BHAC, RAPTOR, BHOSS, ipol.	
PRESENT ACADEMIC STATUS	TDLI Prized Postdoctoral Fellow , Tsung-Dao Lee Institute, Shanghai Jiao-Tong University ◇ Research Topic: General relativistic magnetically driven flows from thin accretion disk. ◇ Mentor: Prof. Yosuke MIZUNO	08/2022-
LAST ACADEMIC STATUS	Academic Visitor , International Centre for Theoretical Sciences-Tata Institute of Fundamental Research ◇ Invited by: Prof. Pallavi BHAT	04/2022-07/2022

	Postdoctoral Fellow, IIT Indore Disciple of Astronomy, Astrophysics and Space Engineering ◇ Research Topic: General relativistic magnetically driven flows from thin accretion disk. ◇ Mentor: Prof. Bhargav VAIDYA	03/2020-03/2022
DOCTOR OF PHILOSOPHY	Department of Physics, IIT Guwahati ◇ Research Topic: Relativistic advective accretion flow around black hole. ◇ Supervisor: Prof. Santabrata DAS	2014-2020
WORK EXPERIENCE	M.Sc. Project, IIT Guwahati Department of Physics ◇ Project Title : Mass ejection from accretion disk around a black hole. ◇ Supervisor: Prof. Santabrata DAS	2013-2014
EDUCATION	IIT Guwahati ,Guwahati,Assam, India ◇ M.Sc. in Physics ◇ CGPA: 8.89/10	2012 - 2014
	DHSK College ,Dibrugarh, Assam, India ◇ B.Sc. in Physics ◇ Percentage: 82.33%	2009 - 2012
Awards Fellowships Grants	◇ Overseas Talent Introduction Program in 2023, China ◇ TDLI prize postdoctoral fellow, China ◇ Best thesis award, Department of Physics, IIT Guwahati, India ◇ Post-Doctoral Fellow, Max Planck Partner group, DAASE, IIT Indore, India ◇ Junior Research Fellowship (JRF), IIT Guwahati, India ◇ Senior Research Fellowship (SRF), IIT Guwahati, India	1/2023- 4/2022- 2020 3/2020 -3/2022 2014 - 2016 2016 - 2019
Research Grants	◇ The National Key R&D Program of China (No. 2023YFE0101200) as participant: 600,000 RMB ◇ Shanghai Municipality orientation program of Basic Research for International Scientists, (Grant No. 22JC1410600) as participant: 1,000,000 RMB ◇ National Natural Science Foundation of China, General grant, (Grant No. 12273022) as participant: 550,000 RMB ◇ TDLI prize postdoctoral Research grant, China: 50,000 RMB/year	01/07/2023 - 30/06/2026 01/12/2022 - 30/11/2024 01/01/2023 - 31/12/2026 4/2022-
QUALIFIED NATIONAL LEVEL EXAMINATIONS:	◇ GATE 2015, ◇ IIT JAM 2012.	

Publications

1. **Dihingia I. K.**, Uniyal A., Mizuno Y., *The Fate of Transonic Shocks around Black Holes and their Future Astrophysical Implications*, In press (2025) *ApJ* (IF 5.5).
2. Shashank S., Abdikamalov A. B., Liu H., Nosirov A., Bambi C., **Dihingia I. K.**, Mizuno Y., *Measuring black hole spins with X-ray reflection spectroscopy: A GRMHD outlook*, In press (2025) *PhysRevD* (IF 5.3).
3. Event Horizon Telescope Collaboration et al. 2025 (+**Indu K. Dihingia**), *Probing jet base emission of M87* with the 2021 Event Horizon Telescope observations*, In press (2025) *A&A* (IF 6.24).
4. Jiang H.-X., Mizuno Y., Lai D., **Dihingia I. K.**, Fromm C. M., *Magnetically Driven Retrograde Precession in Misaligned Black Hole Accretion Flows*, 995, 112 (2025) *ApJ* (IF 5.5).
5. Uniyal A., **Dihingia I. K.**, Mizuno Y., Rezzolla L., *The future ability to test theories of gravity with black-hole shadows*, 993, 97 (2025) *NatAs* (IF 14.3).
6. Uniyal A., **Dihingia I. K.**, Mizuno Y., Kluźniak W., *GRMHD Study of Accretion onto Time-like Naked Singularities*, 993, 97 (2025) *ApJ* (IF 5.5).
7. Jiang H.-X., Mizuno Y., **Dihingia I. K.**, Yuan F., Lin X., Fromm C. M., Nathanail A., et al., *The Physical Origin and Time Lag of Multifrequency Flares from Sgr A**, 990, 81 (2025) *ApJ* (IF 5.5).
8. **Dihingia I. K.**, Uniyal A., Mizuno Y., *Imprints of Different Types of Low-angular-momentum Accretion Flow Solutions in General Relativistic Hydrodynamic Simulations*, 990, 35 (2025) *ApJ* (IF 5.5).
9. Mao J., **Dihingia I. K.**, Mizuno Y., Nagataki S., *Low-angular-momentum Black Hole Accretion: First General Relativistic Magnetohydrodynamic Evidence of Standing Shocks*, 990, 12 (2025) *ApJ* (IF 5.5).
10. Event Horizon Telescope Collaboration et al. 2025 (+**Indu K. Dihingia**), *Origin of the ring ellipticity in the black hole images of M87**, 699, A279 (2025) *A&A* (IF 6.24).
11. Event Horizon Telescope Collaboration et al. 2025 (+**Indu K. Dihingia**), *First polarization study of the M87 jet and active galactic nuclei at submillimeter wavelengths with ALMA*, 699, A265 (2025) *A&A* (IF 6.24).
12. Sarkar B., **Dihingia I. K.**, Misra R., *Thermal conduction and thermal-driven winds in magnetized viscous accretion disk dynamics*, 118, 102377 (2025) *NewA* (IF 1.9).
13. **Dihingia I. K.**, Mizuno Y., *Emergence of cHz Quasiperiodic Oscillations from a Low-angular-momentum Flow onto a Supermassive Black Hole*, 982, L21 (2025) *ApJL* (IF 11.7).
14. Event Horizon Telescope Collaboration et al. 2025 (+**Indu K. Dihingia**), *A multifrequency study of sub-parsec jets with the Event Horizon Telescope*, 695, A233 (2025) *A&A* (IF 6.24).
15. **Dihingia I. K.**, Mizuno Y., Fromm C. M., Younsi Z., *Impact of radiative cooling on the magnetised geometrically thin accretion disk around Kerr black hole*, 2025, *JCAP* (IF 7.28).

16. Moriyama K., Cruz-Orsorio A., Mizuno Y., **Dihingia I. K.**, Uniyal A., *Black hole accretion and radiation variability in GRMHD simulations with Rezzolla-Zhidenko spacetime*, 694, A135 (2025) A&A (**IF 6.24**).
17. Event Horizon Telescope Collaboration et al. 2025 (**+Indu K. Dihingia**), *The persistent shadow of the supermassive black hole of M87 II. Model comparisons and theoretical interpretations*, 693, A265 (2025) A&A (**IF 6.24**).
18. **Dihingia I. K.**, Uniyal A., Mizuno Y., *Distinguishability of a Naked Singularity from a Black Hole in Dynamics and Radiative Signatures*, 978, 1 (2025) ApJ (**IF 5.5**).
19. Event Horizon Telescope Collaboration et al. 2024 (**+Indu K. Dihingia**), *The putative center in NGC 1052*, 692, A205 (2024) A&A (**IF 6.24**).
20. Event Horizon Telescope Collaboration et al. 2024 (**+Indu K. Dihingia**), *Broadband multi-wavelength properties of M87 during the 2018 EHT campaign including a very high energy flaring episode*, 692, A140 (2024) A&A (**IF 6.24**).
21. Event Horizon Telescope Collaboration et al. 2024 (**+Indu K. Dihingia**), *First Very Long Baseline Interferometry Detections at 870 μ m*, 168, 130 (2024) AJ (**IF 5.1**).
22. Uniyal A., **Dihingia I. K.**, Mizuno Y., *Revisited equilibrium solution of Fishbone and Moncrief torus for extended GRMHD simulations*, 970, 172 (2024) ApJ (**IF 5.5**).
23. Jiang H.-X., Mizuno Y., **Dihingia I. K.**, Nathanail A., Younsi Z., Fromm C. M., *Dynamics and Emission Properties of Flux Ropes from Two-Temperature GRMHD Simulations with Multiple Magnetic Loops*, 688, 82 (2024) A&A (**IF 6.24**).
24. Jiang H.-X., **Dihingia I. K.**, Cheng L., Mizuno Y., Zhu T., *Enhanced Blandford–Znajek Jet in Loop Quantum Black Hole*, 2024, 101 (2024) JCAP (**IF 7.28**).
25. **Dihingia I. K.**, Mizuno Y., *Dynamical properties of magnetized low angular momentum accretion flow around a Kerr black hole*, 967, 4 (2024) ApJ (**IF 5.5**).
26. Event Horizon Telescope Collaboration et al. 2024 (**+Indu K. Dihingia**), *First Sagittarius A* Event Horizon Telescope Results. VIII. Physical interpretation of the polarized ring*, 964, L26 (2024) ApJL (**IF 7.8**).
27. Event Horizon Telescope Collaboration et al. 2024 (**+Indu K. Dihingia**), *First Sagittarius A* Event Horizon Telescope Results. VII. Polarization of the Ring*, 964, L25 (2024) ApJL (**IF 7.8**).
28. Jiang H.-X., Liu C., **Dihingia I. K.**, Mizuno Y., Xu H., Zhu T., Wu Q., *Shadows of loop quantum black holes: semi-analytical simulations of loop quantum gravity effects on Sagittarius A* and M87**, 01, 059 (2024) JCAP (**IF 7.28**).
29. Event Horizon Telescope Collaboration et al. 2024 (**+Indu K. Dihingia**), *The persistent shadow of the supermassive black hole of M87. I. Observations, calibration, imaging, and analysis*, 681, A79 (2024) A&A (**IF 6.24**).
30. **Dihingia I. K.**, Mizuno Y., Sharma P., *High-soft to Low-hard State Transition in Black Hole X-Ray Binaries with GRMHD Simulations*, 958, 105 (2023) ApJ (**IF 5.5**).
31. Acharya S., Vaidya B., **Dihingia I. K.**, Agarwal S., Shukla A., *A numerical study on the role of instabilities on multi-wavelength variability in Blazar jets*, 671, A161 (2023) A&A (**IF 6.2**).
32. **Dihingia I. K.**, Mizuno Y., Fromm C. M., Rezzolla L., *Temperature properties in magnetized and radiatively cooled two-temperature accretion flows onto a black hole*, 518, 405 (2023) MNRAS (**IF 5.235**).

33. **Dihingia I. K.**, Vaidya B., Fendt C., *Truncated accretion disc as a possible source of variabilities in BH-XRBs*, 517, 5032 (2023) *MNRAS* (**IF 5.235**).
34. Mitra S., Maity D., **Dihingia I. K.**, Das S., *Study of general relativistic magnetohydrodynamic accretion flow around black holes*, 516, 5092 (2022) *MNRAS* (**IF 5.235**).
35. Das S., Nandi A., Stalin C. S., Rakshit S., **Dihingia I. K.**, Singh S., Aktar R., et al., *On the origin of core radio emissions from black hole sources in the realm of relativistic shocked accretion flow*, 514, 1940 (2022) *MNRAS* (**IF 5.235**).
36. **Dihingia I. K.**, Vaidya B., *Properties of accretion disc, jet, and disc-wind around Kerr black hole*, 43, 23 (2022) *JApA* (**IF 1.766**).
37. Das S., Nandi A., Agrawal V. K., **Dihingia I. K.**, Majumder S., *Relativistic viscous accretion flow model for ULX sources: a case study for IC 342 X-1*, 507, 2777 (2021) *MNRAS* (**IF 5.235**).
38. **Dihingia I. K.**, Vaidya B., Fendt C., *Jets, disc-winds, and oscillations in general relativistic, magnetically driven flows around black holes*, 505, 3596 (2021) *MNRAS* (**IF 5.235**).
39. **Dihingia I. K.**, Maity D., Chakrabarti S., Das S., *Study of relativistic accretion flow in Kerr-Taub-NUT spacetime*, 102, 023012 (2020) *PhysRevD* (**IF 5.407**).
40. **Dihingia I. K.**, Das S., Prabhakar G., Mandal S., *Properties of two-temperature magnetized advective accretion flow around rotating black hole*, 496, 3043 (2020) *MNRAS* (**IF 5.235**).
41. Dihingia I., Das S., Maity D., Nandi A., *Shocks in relativistic viscous accretion flows around Kerr black holes*, 488, 2412 (2019) *MNRAS* (**IF 5.235**).
42. **Dihingia I. K.**, Das S., Nandi A., *Low angular momentum relativistic hot accretion flow around Kerr black holes with variable adiabatic index*, 484, 3209 (2019) *MNRAS* (**IF 5.235**).
43. **Dihingia I. K.**, Das S., Maity D., Chakrabarti S., *Limitations of the pseudo-Newtonian approach in studying the accretion flow around a Kerr black hole*, 98, 083004 (2018) *PhysRevD* (**IF 5.407**).
44. **Dihingia I. K.**, Das S., Mandal S., *Properties of two-temperature dissipative accretion flow around black holes*, 475, 2164 (2018) *MNRAS* (**IF 5.235**).
45. **Dihingia I. K.**, Das S., Mandal S., *A comparative study of single-temperature and two-temperature accretion flows around black holes*, 39, 6 (2018) *JApA* (**IF 1.766**).
46. **Dihingia I. K.**, Das S., Mandal S., *Shocks in two-temperature advective accretion flow around black holes*, 12, 139 (2014) *ASInC*.

Book Chapter

1. **Dihingia I. K.** and Fendt C., *Thin Accretion disks in GR-MHD simulations*, **Springer Singapore, December 24** in “**New Frontiers in GRMHD Simulations**” [ISBN 978-981-97-8521-6], Editors: Cosimo Bambi, Yosuke Mizuno, Swarnim Shashank, and Feng Yuan.

Conferences Talks

- [1] *China-India-Thailand Workshop on High Energy Astrophysics*, **Suranaree University of Technology, Nakhon Ratchasima, Thailand**, October 3-5 December 2025.
- [2] *2025 Postdoctoral Frontier Symposium in Physics and Astronomy*, **Tsung-Dao Lee Institute, Shanghai, China**, September 24–26, 2025.
- [3] *Mini-Workshop Testing General Relativity with Black Holes*, **Fudan University, Shanghai, China**, October 26 April 2025.
- [4] *Transient Phenomena and Physical Processes Around Supermassive Black Holes*, **Tsung-Dao Lee Institute, Shanghai Jiao Tong University**, October 15 to 18, 2024.
- [5] *5th China-India Workshop on High Energy Astrophysics*, **Fudan University, Online mode**, 16-20 December 2024.
- [6] *High-energy Astrophysics/Compact Object Mini-Workshop*, **Tsung-Dao Lee Institute, Shanghai Jiao Tong University**, May 16 to 17, 2024.
- [7] *The 32nd Texas Symposium on Relativistic Astrophysics*, **Tsung-Dao Lee Institute, Shanghai Jiao Tong University**, December 11 to 15, 2023.
- [8] *New Horizons and Singularities in Gravity (ICGC 2023)*, **IIT Guwahati, India**, December 6 to 9, 2023.
- [9] *Advances in Relativistic Astrophysics (AReA)*, 2024, **ARIES, Nainital, India**, Nov 2-4, 2023.
- [10] *Astrophysical Black Holes: A Rapidly Moving Field*, **The University of Hong Kong**, 23–26 Jun 2023.
- [11] *Challenges and Innovations in Computational Astrophysics IV*, **Virtual conference**, 21-23 November 2022
- [12] *North-East Meet of Astronomers (NEMA-VII)*, **Rajiv Gandhi University (RGU)**, Arunachal Pradesh, 27 to 29 January, 2022
- [13] *HEASA 2021*, **South Africa/Johannesburg**, 13-17 September, 2021
- [14] *The 15th Greek Astronomical Conference*, **Patras, Greece**, July 5 to 8, 2021.
- [15] *Extragalactic jets on all scales - launching, propagation, termination*, **An ONLINE conference, jointly organized by MPIA, Germany and IIT Indore, India**, June 14-18, 2021.
- [16] *Astrophysical jets and observational facilities: National perspective*, **ARIES Nainital, India**, 05 - 09 April 2021.
- [17] *XXXIXth meeting of the Astronomical Society of India*, **Hosted jointly by ICTS - TIFR Bengaluru, IISER Mohali, IIT Indore and IUCAA Pune**, 18 - 23 February 2021.
- [18] *43nd cospar scientific assembly*, **COSPAR**, Sydney, Australia, 28 January-4 February 2021.
- [19] *North-East Meet of Astronomers (NEMA) - V*, **Tezpur University**, Tezpur, India 11 - 13 September, 2019.
- [20] *National conference on REcent Trends in the study of Compact Objects:Theory and Observation (RETCO-IV)*, **IUCAA**, Pune, India 17 - 20 April, 2019.
- [21] *International Conference on Computational Mathematics in Nanoelectronics and Astrophysics*, **Indian Institute of Technology Indore**, Indore, India November 1-3, 2018.

	[22] <i>National conference on REcent Trends in the study of Compact Objects:Theory and Observation (RETCO-III)</i>, Indian Institute of Space Science and Technology (IIST) , Thiruvananthapuram, India June 5-7, 2017. [23] <i>The 29th meeting of the Indian Association for General Relativity and Gravitation (IA-GRG)</i>, Department of Physics, IIT Guwahati, Guwahati, India May 18-20, 2017. [24] <i>The conference, Wide Band Spectral and Timing Studies of Cosmic X-ray Sources</i>, TIFR, Mumbai, India 13 February, 2016. [25] <i>One day symposium on 100 years of General Relativity: Where do we stand?</i>, Department of Physics, IIT Guwahati , Guwahati, India January 10-13, 2017. [26] <i>National conference on REcent Trends in the study of Compact Objects:Theory and Observation (RETCO-II)</i>, Aryabhata Research Institute of Observational Sciences , Nainital, India May 06 - 08, 2015.
Invited Talks	[1] <i>Unveiling accretion flows around our galaxy's supermassive black hole</i>, Bar-Ilan University, Ramat Gan, Israel, 13th November, 2024 [2] <i>Unveiling accretion flows around our galaxy's supermassive black hole</i>, South-Western Institute for Astronomy Research, Yunnan University, Kunming, China, 11th September, 2024 [3] <i>Unveiling accretion flows around our galaxy's supermassive black hole</i>, Yunnan Astronomical Observatory, Kunming, China, 9th September, 2024 [4] <i>Unveiling accretion flows around our galaxy's supermassive black hole</i>, IIT Indore, India, 8th July, 2024 [5] <i>Understanding physics around Black Holes with General Relativistic Magneto Hydrodynamics (GRMHD)</i>, IIT Guwahati, India, 28 Nov, 2023 [6] <i>Understanding physics around Black Holes with General Relativistic Magnetohydrodynamics (GRMHD)</i>, Guwahati University, 4 Dec, 2023. [7] <i>Understanding physics around Black Holes with General Relativistic Magnetohydrodynamics (GRMHD)</i>, Tezpur University, 20 Nov, 2023. [8] <i>Exploring Accretion Properties Around Black Holes with GRMHD</i>, Fudan University, Sep 11, 2023. [9] <i>Understanding flow around black holes with GRMHD simulations</i>, IISC, Bangalore, 25 July, 2022. [10] <i>Understanding flow around black holes with GRMHD simulations</i>, ICTS-TIFR, 24 June 2022.
Teaching Experience	I have experience working as a teaching assistant for M.Sc. and B. Tech. courses during my Ph.D. at IIT Guwahati, India. I assisted in Numerical labs, Basic Physics lab, and Electronics lab. I have been training grad and undergrad students for scientific research during my Ph.D. and postdoctoral tenure. Currently, I am guiding three Ph.D. students at TDLI.
DECLARATION	I hereby declare that all above-mentioned information is in accordance with fact or truth up to my knowledge and I bear the responsibilities for the correctness of the above mentioned particulars.